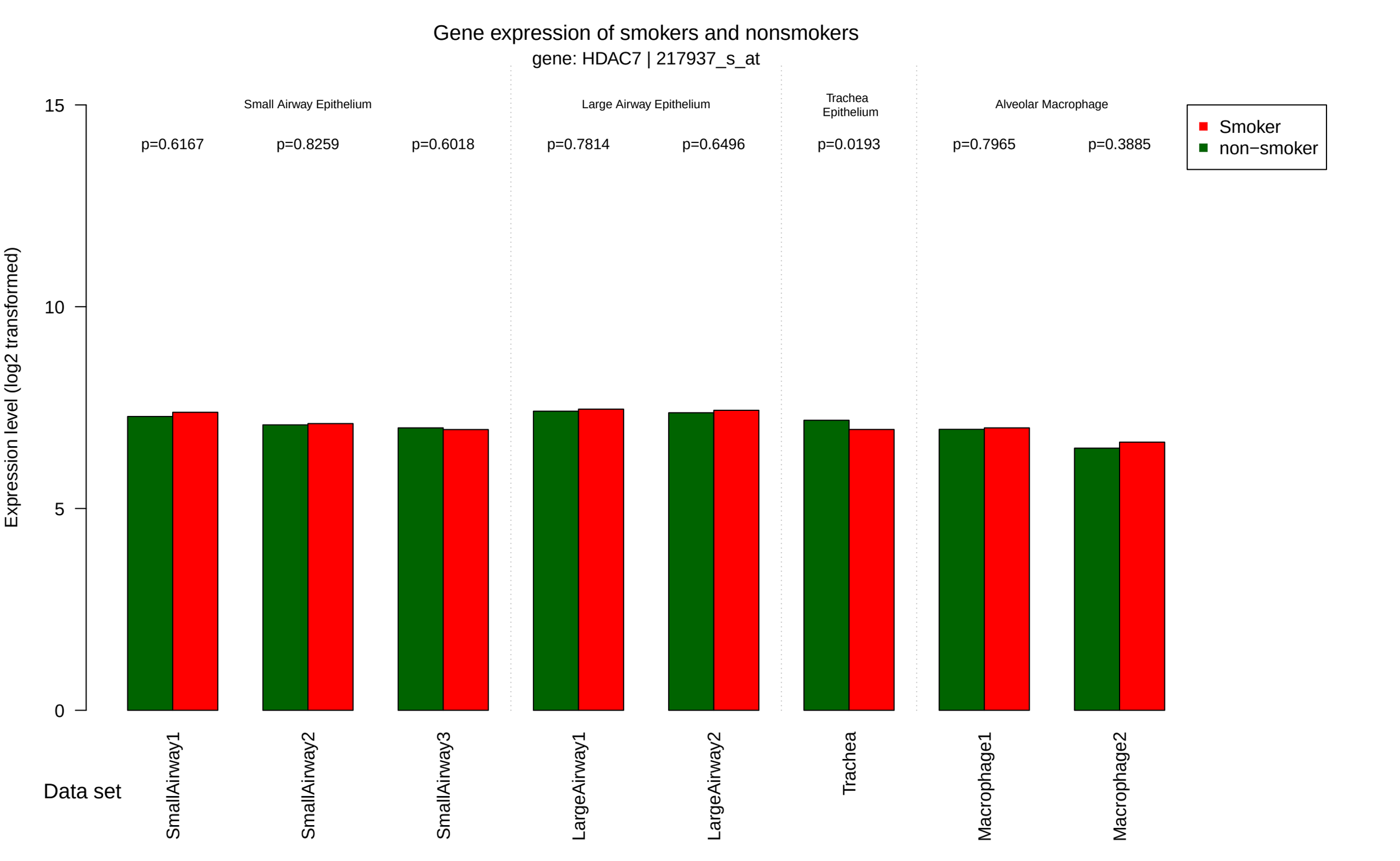
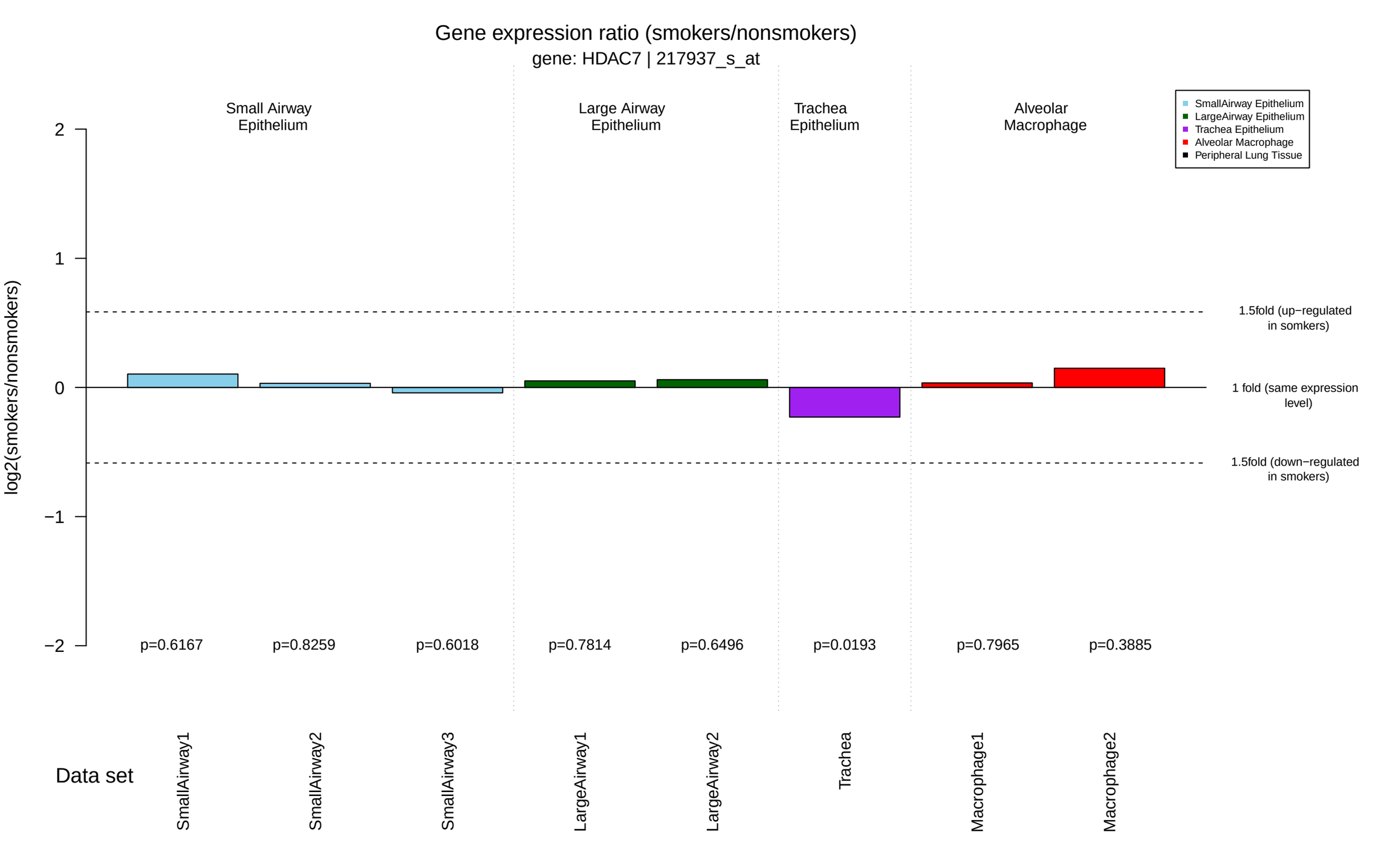
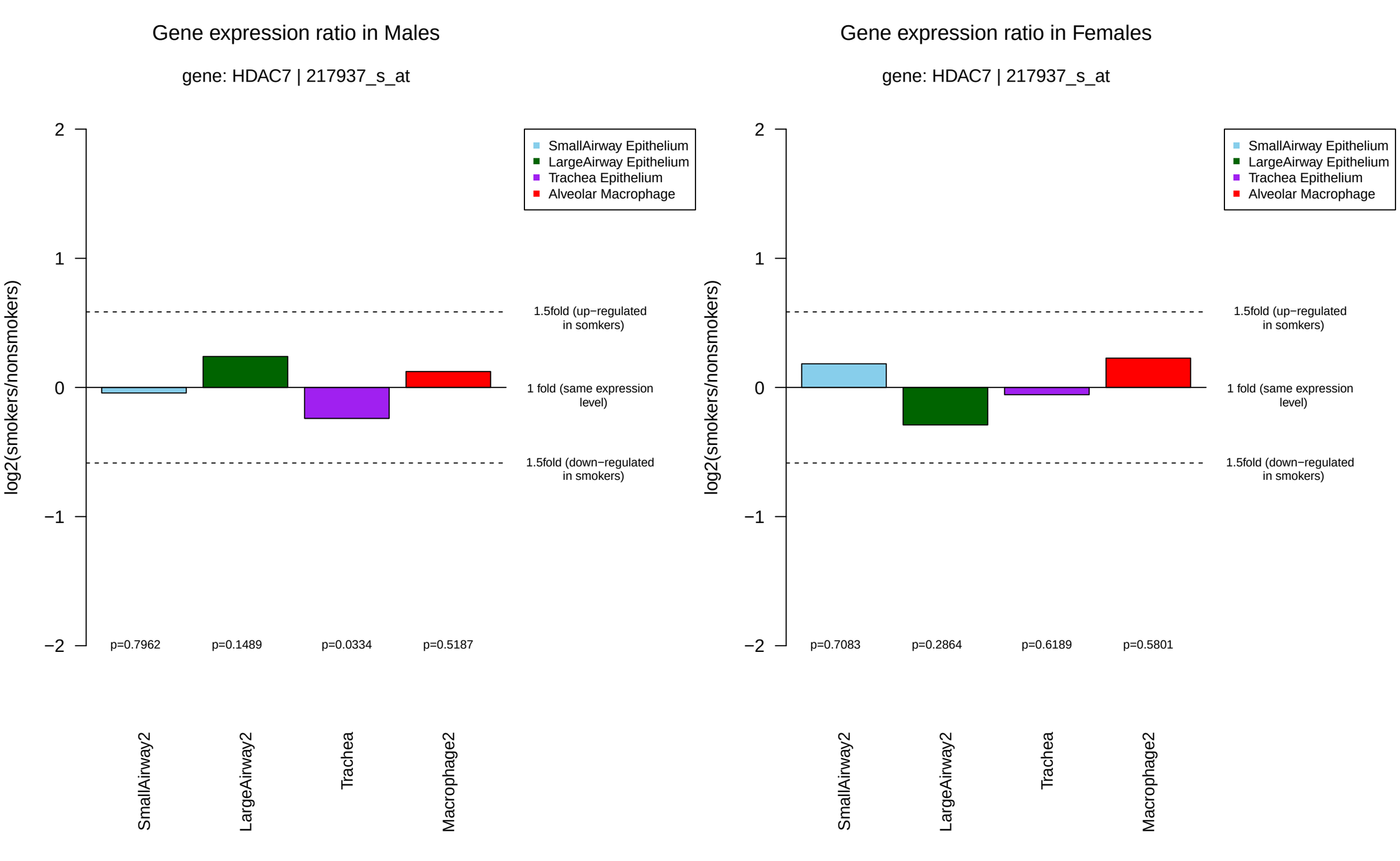
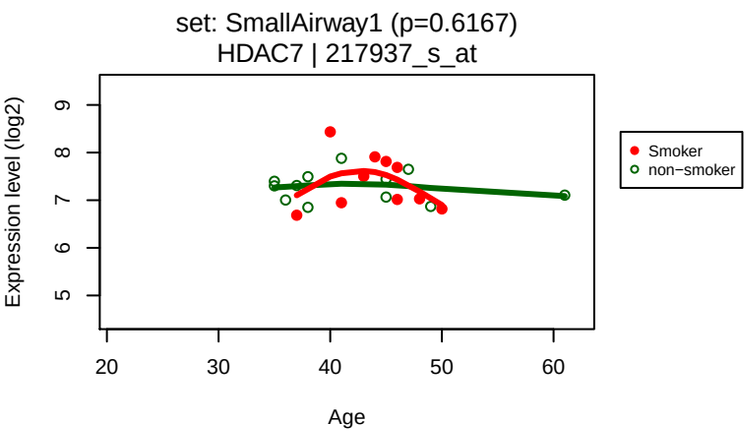
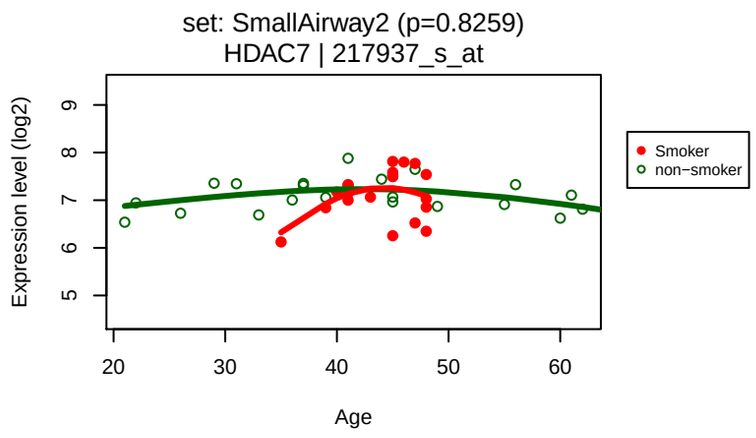
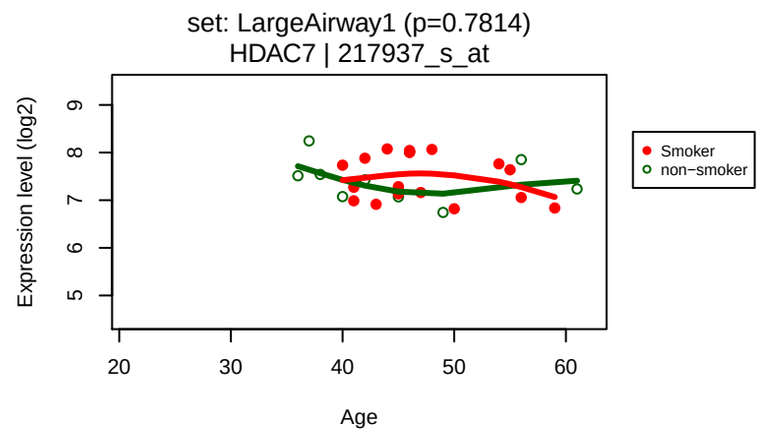

Gene expression of smokers and nonsmokers
gene: HDAC7 | 217937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6167
p=0.8259
p=0.6018
p=0.7814
p=0.6496
p=0.0193
p=0.7965
p=0.3885
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

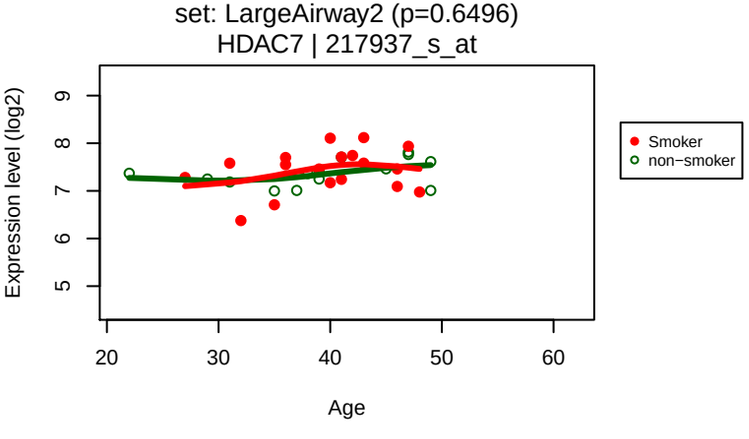

Gene expression ratio (smokers/nonsmokers)
gene: HDAC7 | 217937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6167
p=0.8259
p=0.6018
p=0.7814
p=0.6496
p=0.0193
p=0.7965
p=0.3885
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

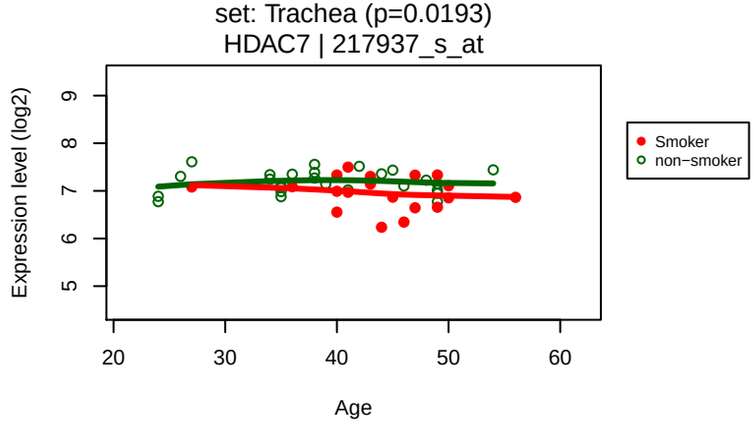

Gene expression ratio in Males
gene: HDAC7 | 217937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7962
p=0.1489
p=0.0334
p=0.5187
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC7 | 217937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7083
p=0.2864
p=0.6189
p=0.5801
SmallAirway2
LargeAirway2
Trachea
Macrophage2

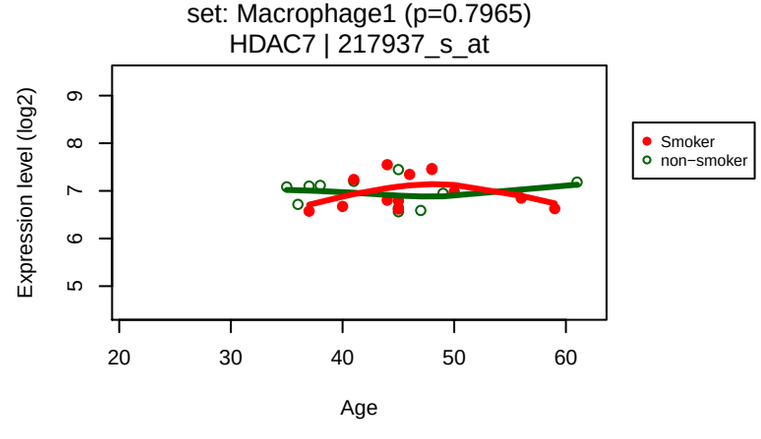

set: SmallAirway1 (p=0.6167)
HDAC7 | 217937_s_at
Expression level (log2)
Age
Smoker
non−smoker

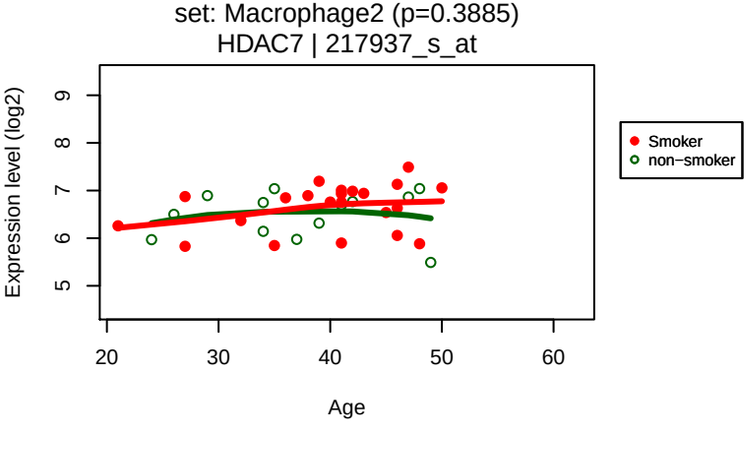

set: SmallAirway2 (p=0.8259)
HDAC7 | 217937_s_at
Expression level (log2)
Age
Smoker
non−smoker

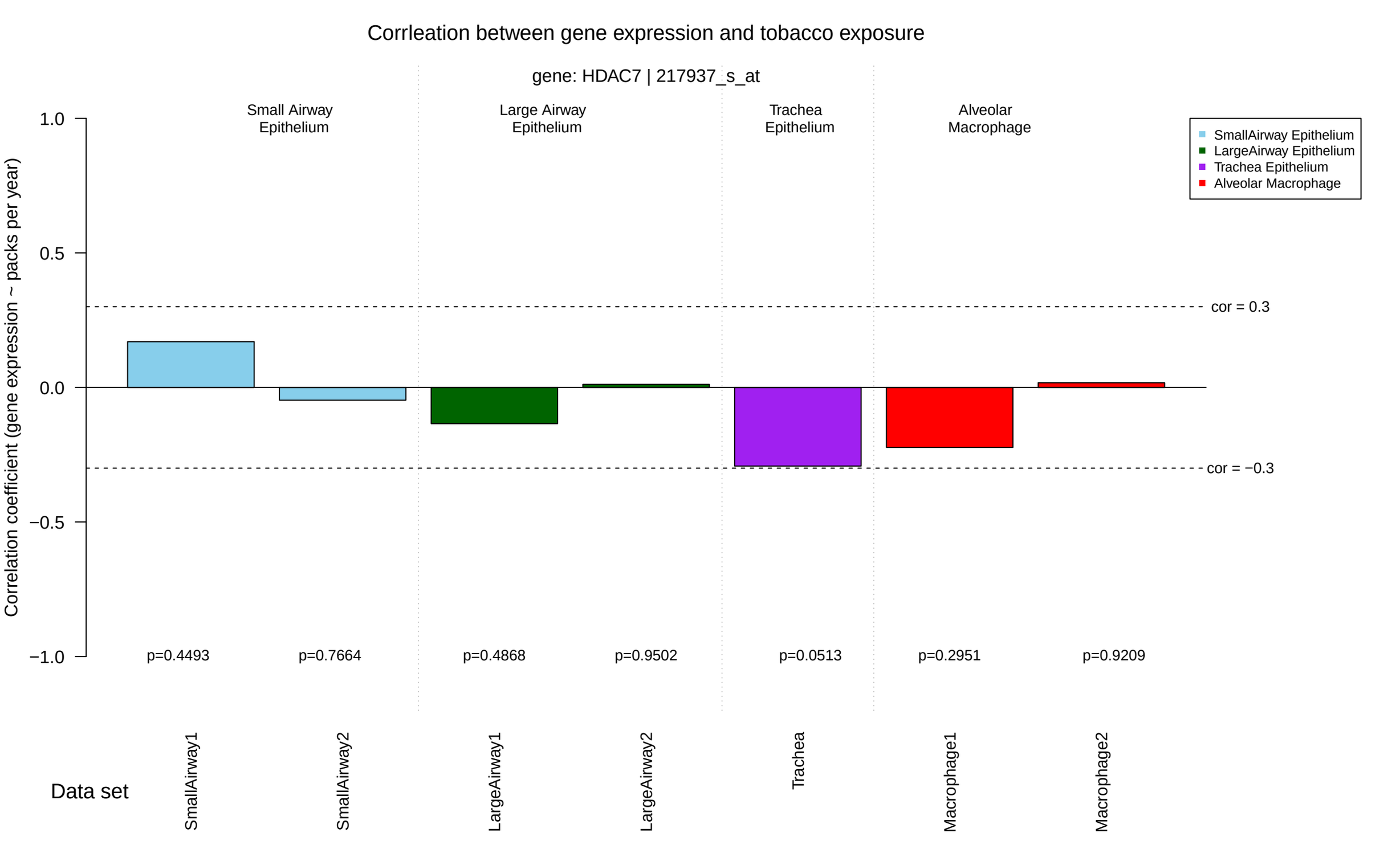

set: LargeAirway1 (p=0.7814)
HDAC7 | 217937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6496)
HDAC7 | 217937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0193)
HDAC7 | 217937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7965)
HDAC7 | 217937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3885)
HDAC7 | 217937_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC7 | 217937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4493
p=0.7664
p=0.4868
p=0.9502
p=0.0513
p=0.2951
p=0.9209
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2